

DRY-RUN GUIDE | temp.java
Source: java\1st_belt\temp.java

CORE BEHAVIOR

convertTemperature converts Fahrenheit to Celsius for unit F and Celsius to Fahrenheit for C; other unit characters produce no output.

REPRESENTATIVE INPUT / CALL

Input: '100 C'.

STEP TRACE

C branch computes $100 * 9 / 5 + 32 = 212$; printf formats two decimal places.

EXPECTED OUTPUT / RETURN

'212.00'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Challenge #1
003 |
004 | Write a program that converts a temperature from Celsius to Fahrenheit or vice
    | versa.
005 |
006 | You need to convert a temperature based on the unit provided by the user (either
    | Celsius or Fahrenheit).
007 | The program should ask the user to specify the temperature unit (Celsius or
    | Fahrenheit) and then perform the conversion.
008 |
009 | Formulas:
010 | Celsius to Fahrenheit:  $F = (C * 9 / 5) + 32$ 
011 | Fahrenheit to Celsius:  $C = (F - 32) * 5 / 9$ 
012 |
013 | Input Format:
014 | The first line of input contains a floating point number for temperature.
015 | The second line of input contains the unit of the input temperature ('C' for Celsius
    | or 'F' for Fahrenheit).
016 |
017 | Output Format:
018 | The program will output the converted temperature rounded to two decimal places.
019 | */
020 |
021 | import java.util.Scanner;
022 |
023 | public class Solution {
024 |     public static void convertTemperature(float temp, char unit) {
025 |         // Implement logic here
026 |         if (unit == 'F') {
027 |             float result = (temp - 32) * 5.0f / 9.0f;
028 |             System.out.printf("%.2f\n", result);
029 |         }
030 |         else if (unit == 'C') {
031 |             float result = (temp * 9.0f / 5.0f) + 32;
032 |             System.out.printf("%.2f\n", result);
033 |         }
    |     }
```

```
034 |     }
035 |
036 |     public static void main(String[] args) {
037 |         // Call the function after taking input
038 |         Scanner scanner = new Scanner(System.in);
039 |         float temp = scanner.nextFloat();
040 |         char unit = scanner.next().charAt(0);
041 |         convertTemperature(temp, unit);
042 |     }
043 | }
044 |
```